# Product Requirement Document (PRD)

## Local Routes Confidence Layer (LRCL)

## I. Executive Summary

Commuters in major Indian tier-1 cities like Bengaluru, Mumbai, and Hyderabad face severe traffic congestion because navigation platforms consistently direct vehicle traffic onto identical primary arterial roads. While local drivers successfully utilize sub-arterial corridors and shortcuts based on localized knowledge, standard navigation engines avoid these routes due to data asymmetry and a lack of verifiable trust regarding lane width, structural obstructions, and overall safety".

The Local Routes Confidence Layer (LRCL) addresses this infrastructure gap. By analyzing crowd-sourced telemetry and implementing human-verified validation loops, the platform identifies, scores, and safely utilizes high-utility local roads to distribute traffic dynamically, bypass gridlock, and unlock hidden urban transit capacity.

## II. Problem Statement and Market Friction

- ***The Last-Mile and Arterial Paradox*:** Despite continuous capital investments in mass transit networks (Metros, BRTS), commuter transit to final destinations heavily relies on a limited number of primary arterial roads, leading to systemic infrastructure failure during peak commuting hours.

- ***The Map's Blindspot*:** Global mapping algorithms deliberately exclude unclassified or narrow residential lanes because their systems cannot accurately verify vehicle-type compatibility, real-time road quality, or localized safety conditions.

- ***Information Asymmetry*:** A significant disconnect exists between the experiential "mental map" possessed by local residents and the static data layer of "digital maps." This data gap forces non-local drivers onto congested main roads, compounding stress on main arterial roads.

## III. Product Vision & Objectives

| Targets | Rationale |
|---|---|
| Map 10,000 to 50,000 high-utility shortcuts across target cities. | 1. A tier-1 city like Bengaluru features roughly 12,000 kilometers of total road network, but severe gridlock is highly concentrated within specific commercial and technology corridors.<br>2. Isolating the top 10,000 high-utility shortcuts is sufficient to optimize these high-friction zones during the pilot phase.<br>3. Scaling the database to 50,000 segments ensures comprehensive coverage across multi-city rollouts during full launch |

| Targets | Rationale |
|---|---|
| Divert 15% to 18% of platform traffic away from primary arterial roads. | 1. Traffic congestion doesn't build up linearly, rather it explodes all at once.<br>2. When a primary highway reaches its breaking point, complete clearance is unnecessary to restore flow.<br>3. Diverting just 15% to 18% of eligible platform vehicles (specifically highly agile two-wheelers and auto-rickshaws) onto parallel local lanes effectively makes the main road flow smooth. |
| Reduce absolute commute times by 8 to 12 minutes per peak-hour trip. | 1. The baseline peak-hour commute in target Indian metros averages 45 to 60 minutes.<br>2. A verified, structurally optimized local shortcut historically reduces trip duration by approximately 20%, which mathematically corresponds to an absolute savings of 8 to 12 minutes for the end-user. |

## IV. Feature Architecture and Functional Requirements

1. **Phase 1: Automated Discovery & Confidence Scoring**

   The platform ingests passive, anonymized GPS telemetry to identify geographic coordinates where users consistently navigate outside the officially classified road network.

   a. **Segment Flagging**: Automatically identify and isolate unmapped or unclassified road segments that exhibit recurring, high-volume user traversal.

   b. **Confidence Scoring:** Every unverified segment is assigned a dynamic realtime score based on unique user volume per hour, and localized velocity patterns.

   c. **Vehicle Modality Classification**: Analyze velocity profiles, deceleration rates, and historical lane footprints to programmatically differentiate whether a segment safely accommodates two-wheelers, three-wheelers (Autos), or standard four-wheelers.

   d. **Segment Consistency Auditing:** Evaluate dwell-time metrics and continuous motion data to guarantee the flagged path represents an active, unobstructed transit corridor rather than a dead end, private driveway, or static parking zone.

2. **Phase 2: Conditional Routing Logic (The Trust Threshold)**

   To maintain user trust and avoid routing vehicles into volatile or hazardous environments, the navigation engine will suggest an unverified local shortcut only when the following conditional gates are met simultaneously:

| Conditions | Rationale |
| --- | --- |
| It saved at least 20% of ETA | 1. Navigating narrow residential lanes increases cognitive load and driver stress.<br>2. If a shortcut saves negligible time, users feel penalized for taking the risk.<br>3. The route must yield a minimum 20% time savings to cross the human friction threshold and preserve long-term product trust. |
| Arterial Delay is more than 40% of the normal time. | 1. Out of civic responsibility to quiet residential neighborhoods, the platform must not divert traffic unless necessary.<br>2. The routing engine preserves side streets for local residents until major arteries experience active structural failure |
| Satisfied the time window 06:00 to 22:00. | 1. User safety is the absolute priority. At night, narrow urban side streets present elevated operational risks, including unlit obstacles, open drainage networks, stray animals, and gates closed by neighborhood associations.<br>2. Restricting suggestions to daylight hours protects users and minimizes nighttime residential noise. |

3. **Phase 3: Gamified Community Verification**
   Transitioning a discovered segment from an "Unverified Shortcut" to an "Official Default Route" requires human-in-the-loop validation.

   a. ***Local Guides Incentivization:*** Deploy a targeted incentive framework within the existing Local Guides network. Users earn explicit gamified points, platform rewards, or micro-monetary bonuses for successfully traveling through flagged segments to submit real-time confirmation, point-of-interest photography, and vehicle clearance verification.

   b. ***Graduation Metric:*** A shortcut graduates to default routing status only after achieving a statistically significant threshold of unique user confirmations across diverse vehicle types. This multi-user validation mathematically verifies the route is structurally stable, safe, and universally accessible for public navigation.

## V. Execution Roadmap

a. ***Month 1-2: Data Gathering and Market Research***
   Data science teams will query historical, first-party GPS ping databases to map where users are already navigating off the official network. Because this data is generated natively via our active user base, primary telemetry is accessible immediately without requiring external acquisition costs.

b. ***Month 3-4: Implementing the Routing Logic & Pilot Testing***
Integrate the conditional routing algorithms into a staging environment. Execute a closed beta test using high-frequency commercial users (delivery partners and ride-hailing drivers). This operational sandboxing allows the engineering team to isolate algorithmic edge cases, calibrate safety logic, and rapidly iterate on routing calculations based on real-world feedback.

c. ***Month 5-10: Gamified Incentives & Scaling***
Deploy the localized monetary and point-based incentive structures to the broader Local Guides community to systematically verify remaining unmapped segments. Clear residual software defects, address localized map anomalies, and graduate the feature for generalized public consumer access.

## VI. Key Performance Indicators (KPIs)

The operational health and business success of the LRCL feature rollout will be monitored using the following performance metrics:

1. More than 50,000 unofficial shortcuts (representing a significant density of usable alternative city paths) successfully mapped, scored, and verified with a platform Confidence Score more than 80%.

2. Realize an absolute reduction of 8 to 12 minutes in peak-hour travel times for opt-in consumer cohorts.

3. Achieve more than 75% positive response rate on post-trip micro-surveys asking: "Would you take this local route again?" Negative responses will serve as a primary telemetry signal to downgrade a route's stress score and recalibrate the routing engine.

4. Secure a 15% to 18% reduction in platform traffic due to agile vehicle density across targeted primary arterial corridors during morning and evening peak traffic hours compared to historical seasonal baselines.